1. Marketing Analysis

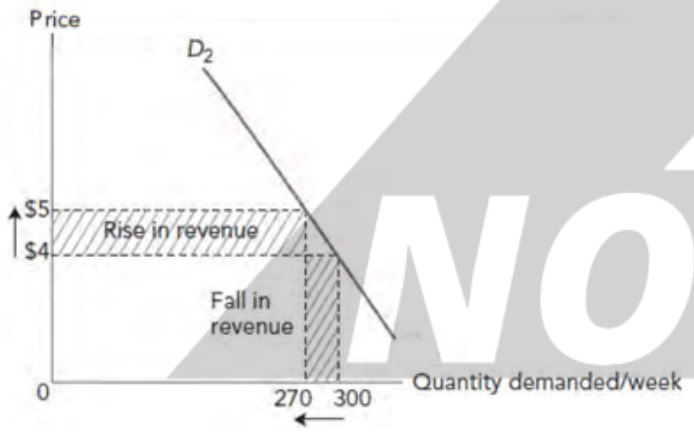
1.1. Elasticity

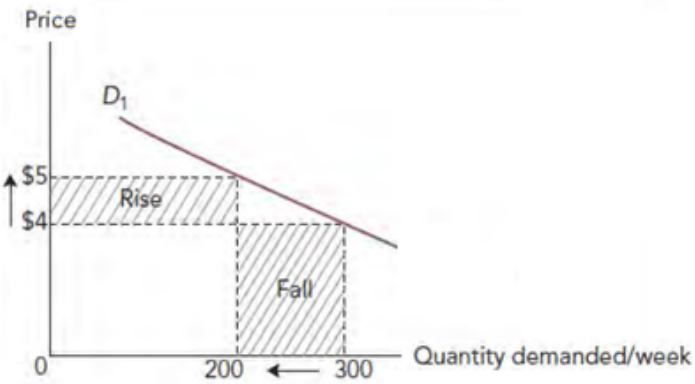
Elasticity of demand

- Marketing decisions on pricing, products, promotions, and distribution should use up-to-date, relevant information. **For example, marketing managers benefit from knowing:**
  - How will a 10% price increase affect demand?
  - How will sales change with a 1 million dollar boost in promotions?
  - How will consumer income changes impact product demand?
- Marketing departments assess how price, promotions, and consumer income affect demand. **Elasticity of demand measures this impact.**

Price elasticity of demand

- The diagrams below show that curve **D2–D2** is steeper than **D1–D1**.
- A steeper curve means less change in demand when prices rise.
- When both products' prices increase equally, demand for product B drops more than for product A.
- Product A's total revenue increases, while product B's falls, as shown by the shaded areas.





- This relationship between price changes and the proportion of the resulting change in demand is known as price elasticity of demand (PED).
- Demand curves show this concept. Product A's demand is less elastic, or responsive, to price changes than product B's. This can also be measured mathematically.
  - **PED =** Percentage change in Quantity Demanded / Percentage change in Price

Interpreting price elasticity results

- PED results should be interpreted carefully.

Value of PED (ignoring minus sign)	Classification	Explanation
Zero	Perfectly inelastic demand	Demand stays the same at any price. No product has this PED in reality.
Between 0 and 1	Inelastic demand	Demand changes less than price. With this PED, higher prices increase revenue, but further increases make demand more elastic.
Unitary	Unit elasticity	When PED = 1, demand changes equally with price, keeping revenue constant. Sales revenue is maximized at this point.
Between 1 and infinity (∞)	Elastic demand	When demand changes more than price, lowering price increases demand and revenue.
Infinity (∞)	Perfectly elastic demand	At one price, demand is infinite but drops to zero with any price increase. No product has this PED in reality.

- Price elasticity of demand (PED) is usually negative due to the inverse relationship between price and demand. A price drop increases demand, while a price rise decreases it. The negative sign is often ignored, focusing instead on the numerical value.
- **Example:** In Figure 21.2, price rose from 4 to 5 dollars, reducing demand from 300 to 270 units. Calculate the PED.
  - **step 1:** calculate percentage change in price
  - **step 2:** calculate percentage change in demand
  - **step 3:** use the formula
    - % change in demand = 10
    - % change in price = 25
    - **PED = $\frac{10}{25}$ = 0.4 (remember, this result is negative)**
- A PED of 0.4 means demand changes by 0.4% for every 1% change in price. Since this is less than one, demand is inelastic. Consumers respond minimally to price changes.
- A price increase raises revenue, while a decrease lowers it due to little change in demand.

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Factors that determine price elasticity of demand

- Factors Affecting PED:
  - **Necessity:** Essential products like salt have inelastic demand as consumers react less to price changes.
  - **Competitors:** More substitutes make demand elastic, while fewer competitors reduce elasticity.
  - **Loyalty:** Strong brands (e.g., Coca-Cola) maintain demand despite price rises, driven by advertising and product differentiation.
  - **Income Proportion:** Low-cost items (e.g., matches) have inelastic demand since price changes minimally impact budgets.

Impact of price elasticity of demand on business decisions

- Impact of PED on Business Decisions:
  - **Pricing:** Raise prices on routes with low PED and lower them on routes with high PED to boost revenue. Supports price discrimination strategies.
  - **Wages:** Wage increases are more feasible if PED is low, as higher prices won't significantly reduce demand. High PED makes passing wage costs to consumers harder.

Income elasticity of demand

- Elasticity applies to variables beyond price. When consumer income changes, demand usually rises. Income elasticity of demand measures how much it changes.
  - **Income elasticity = Percentage change in demand for the product / Percentage change in consumer incomes**
- Example: After a 5% income drop, demand for own-label cereal rose 10%, branded cereal fell 10%, and salt declined 1%. Verify these income elasticity results in the table below.

Own-label cereal	Branded Cereal	Salt
Income elasticity of demand	-2	2

Interpreting income elasticity results

- The own-label cereal is an inferior good with a superior branded alternative. Demand rises as income falls and falls as income rises. Income elasticity is negative and should always show the negative sign.
- The branded product is a normal good with positive income elasticity. Demand rises with income and falls when income drops. If income elasticity exceeds 1, the product is a luxury good with inferior substitutes. Demand changes more than income.
- Salt is a necessity with low positive income elasticity. Demand is income inelastic, remaining stable regardless of income changes.

Impact of income elasticity on business decisions

- **Income elasticity influences business decisions in two ways:**
  - In economic growth, businesses increase output of income-elastic products like luxury goods. New products should have high income elasticity.
  - In a recession, businesses focus on basic, lower-priced products. Firms selling inferior goods may boost output to meet rising demand. New products should be affordable, not exclusive or costly.

Promotional elasticity of demand

- Promotional elasticity of demand is measured by:
  - Promotional elasticity = $\frac{\text{Percentage change in demand for the product}}{\text{Percentage change in promotional spending}}$
- **Increased promotion usually raises demand, resulting in positive promotional elasticity.**
  - If above 1 (like product B), demand rises more than promotional spending, showing elastic demand.
  - If below 1 (like product A), demand is inelastic. Rarely, promotional elasticity can be negative if campaigns offend consumers, causing demand to drop.

Interpreting promotional elasticity of demand

- Increased promotion typically boosts demand, **making promotional elasticity positive.**
- If the result **exceeds 1**, demand rises more than promotion spending **(elastic)**.
- If **below 1**, demand rises less **(inelastic)**.
- Rarely, promotional elasticity **may be negative if offensive campaigns reduce demand.**

Impact of promotional elasticity on business decisions

- Firms should **focus promotion** on products with **high promotional elasticity** and **reduce spending** on those with **low elasticity.**
- However, **low elasticity** may result from **ineffective campaigns** rather than the product itself.
- **Poorly designed promotions** targeting the **wrong audience can limit sales.**
- Before cutting promotion, firms should **investigate the cause of low elasticity**, as better campaigns could drive higher sales.

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Limitations of the measures of elasticity

- Price elasticity of demand (PED) has three key limitations:
  - Assumes no other changes: A price cut may boost sales, but factors like competitor exits or rising incomes can also influence demand, making accurate PED calculation difficult.
  - Quickly outdated: PED changes as consumer tastes shift and new competitors enter. Past calculations may no longer reflect current market conditions.
  - Difficult to calculate: PED often relies on old sales data or market research, which can be inaccurate, especially for new products.
- Income elasticity: Rising incomes may not boost demand if competitors launch better products.
- Promotional elasticity: Sales changes may stem from factors beyond the promotion, like competitor campaigns or economic shifts, making comparisons unreliable. Sales may have risen regardless of promotion due to favorable conditions.
- Despite this, consistent results can provide valuable insights for future marketing decisions.

1.2. Product development

- Fast-changing markets **require constant new product development** to stay competitive.
- Without it, **businesses risk selling outdated products.**
- This is **crucial in consumer electronics, cars, cameras, pharmaceuticals, and industrial machinery**, where innovation drives profits and competitiveness.
- For a new product to succeed, it must:
  - Offer **desirable features** consumers will pay for.
  - Stand out with a **unique selling point**.
  - Be **marketed effectively** to inform consumers.

The process of new product development

- NPD includes seven stages:

1) Generating new ideas: sources of new ideas for product development

- New product ideas come from various sources:
  - R&D Department – Costly, not all businesses have one.
  - Competitor Adaptation – Avoid copyright/patent issues. Similar products may drive consumers to the original.
  - Market Research – Focus groups reveal desired products.
  - Employees – Generate ideas through product/customer interaction. Boosts motivation.
  - Sales Employees – Offer insights from direct consumer contact.
  - Group Brainstorming – Collaborative idea generation surpasses individual efforts.

2) Idea screening

- This stage eliminates ideas unlikely to succeed commercially. Developing and marketing new products is costly, so only viable ideas should move forward.
- Key screening questions include:
  - How will target consumers benefit?
  - Is manufacturing technically feasible?
  - Will it be profitable at the expected price?

3) Concept development and testing

- This stage refines the product idea by evaluating features, costs, and target consumers.
- Key points to consider:
  - Who are the likely consumers?
  - What features and benefits should the product have?
  - How will consumers respond? Use market research to test reactions.
  - What are the most cost-effective manufacturing methods?
  - What will production cost?

4) Business analysis

- This stage evaluates the new product's impact on costs, sales, and profits. Pricing is set based on customer feedback and competitor data. Sales, market share, and break-even levels are estimated.
- Key considerations include:
  - Is financing available for development?
  - Can the product be patented?
  - Does it align with the existing product mix?
  - How might economic changes affect future sales?

5) Product testing

- This stage ensures the product meets technical standards and consumer expectations.
- Testing includes:
  - Developing a prototype
  - Testing in typical-use conditions
  - Gathering feedback from focus groups
  - Adapting the product based on results and customer input

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6) Test marketing

- A small, representative test market is selected to observe consumer behavior and gather feedback before a full launch. Benefits include:
  - Measuring real consumer responses
  - Reducing risk and poor publicity from product failure
  - Addressing product weaknesses based on feedback
- Limitations include high costs and exposure to competitors.
- If results are poor, products may be withdrawn.
- Cheaper alternatives include distributing free samples for consumer feedback.

7) Commercialisation

- This stage is the full-scale product launch, marking the introduction phase of the product life cycle.
- A promotional strategy informs the market, and distribution channels are stocked to ensure availability.
- The first weeks or months are critical, with marketing closely monitoring sales data.

Research and Development

- Effective R&D is vital for maintaining market leadership and competitive advantage. Firms like Apple, Samsung, Toyota, and Honda have boosted performance through R&D.

The importance of R&D

- Inventions drive new product possibilities, some becoming profitable innovations that help businesses grow and stay competitive.
- Innovative products offer unique selling points, allowing premium pricing and higher margins.
- However, R&D is costly and uncertain, with top UK pharmaceutical firms projected to spend over 12 billion dollars in 2020, often on drugs years from market.
- Some businesses avoid R&D, opting to license or copy products, a safer but less imaginative follow-the-leader strategy that risks legal issues.
- R&D strategies vary:
  - Offensive – Lead with innovation to gain market share and dominance, reinvesting profits into more R&D (e.g., Apple).
  - Defensive – Learn from innovators, improving or diversifying products to avoid falling behind.

R&D: an evaluation

- Not all R&D results in breakthroughs. For instance, despite billions spent, the drug industry has not found a cure for the common cold.
- Many inventions also fail to reach the market due to design flaws, manufacturing issues, or competition. Even launched products can fail due to:
  - Inaccurate market research
  - Poor marketing or pricing
  - Obsolete technology
  - Superior competitor products
- Despite these risks, many businesses continue investing in R&D, believing long-term successes will outweigh short-term failures.
- However, short-term pressures to satisfy shareholders can reduce R&D investment.

Factors that influence the level of R&D expenditure by a business

- Several factors influence R&D investment:
  - Industry nature – Rapid technology changes in sectors like pharmaceuticals, defense, and automotive require substantial R&D.
  - Competitor spending – To maintain market share and leadership, businesses often need to match or exceed competitors' R&D budgets.
  - Business expectations – Optimism about economic growth and consumer demand encourages higher R&D spending.
  - Risk profile – A business's willingness to invest long-term, rather than focusing on short-term profits, affects R&D budgets.
  - Government policy – Grants and tax incentives for R&D influence business decisions.

1.3. Sales forecasting

- Accurate sales forecasting reduces business risks, offering benefits like:
  - Operations can plan production, materials, and inventory.
  - Marketing knows how many products to sell and distribute.
  - HR can plan workforce needs with the right skills.
  - Cash flow forecasting becomes more reliable.
- However, 100% accuracy is impossible due to external factors like economic changes, competitor actions, and pricing adjustments.
- Most businesses use sales forecasts to minimize uncertainty in future demand.

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The need to forecast sales

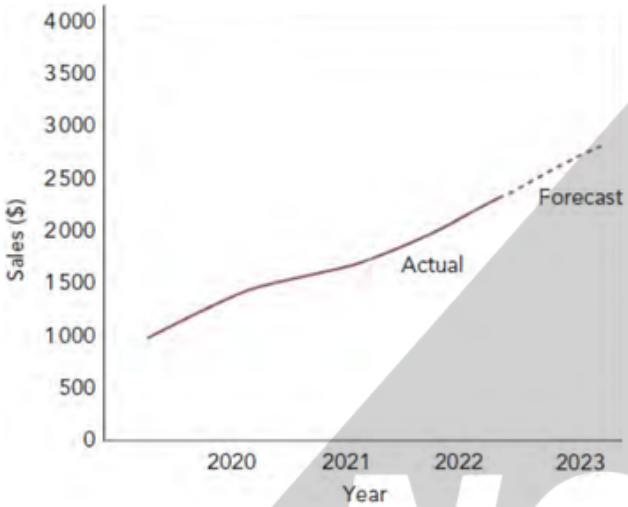
- Sales forecasts are crucial for:
  - Market planning and new product screening.
  - Planning workforce needs.
  - Preparing resources like materials and equipment.
  - Predicting capacity requirements for operations.
  - Informing decisions on price changes and promotions, especially if demand is expected to decline.

Time series analysis

- This sales forecasting method relies on past sales data organized chronologically as a time series.

Extrapolation

- Extrapolation predicts future sales by extending past trends on a time series graph.
- This method assumes stable sales patterns, making it unreliable when conditions change.



Moving averages

- This method is more advanced than graphical extrapolation, identifying factors expected to impact future sales, such as:
  - The trend
  - Seasonal fluctuations
  - Cyclical fluctuations
  - Random fluctations
- Most questions on this topic focus on identifying trends and seasonal fluctuations.
- The moving average method in Table 1 analyzes these factors for ice cream sales, enabling short-term forecasts.

- Points to note
  - The moving average method calculates totals from several sales figures. In Table 2, each total in column 4 represents four quarters, forming a four-quarter moving total. This duration was chosen due to consistent variation over four quarters, such as peak ice cream sales in summer. For daily data, a seven-period total might apply, reflecting weekly variations, while monthly data might use a twelve-period total.
  - Dividing the four-quarter moving total by 4 yields an average that doesn't align with any specific quarter (see Table 1). To fix this, two four-quarter totals are added to form an eight-quarter total (column 5, Table 2), which is then divided by 8 to produce the quarterly moving average (column 6).
  - The quarterly moving average represents the trend, smoothing seasonal fluctuations to reveal the data's underlying movement.
  - Seasonal fluctuations can then be calculated as the difference between actual sales and the trend, as shown in Table 2.
- Seasonal variation (column 7) = actual result (sales revenue, column 3) – moving average trend (column
- Ensure the correct sign for your results. A negative value indicates that quarterly sales are typically below the trend due to seasonal factors.
  - The average seasonal variation (ASV) is found by adding the seasonal variations for each quarter and dividing by the number of results.
    - 43.75 + 52.5 + 58.75 = 155
  - The average seasonal variation for Q3 = 155/3 = ASV 51.67.

Quarter	Sales	Four-quarter total	Four-quarter average
1	20		
2	30	27.5	
3	50		
4	10	110/4	

Table 1

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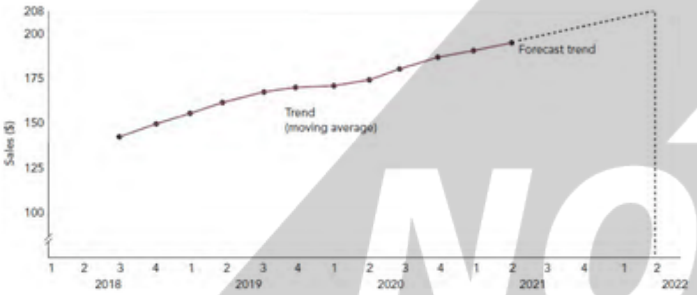
1	2	3	4	5	6	7	8
Year	Quarter	Sales revenue	Four-quarter moving total	Eight-quarter moving total	Quarterly moving average (Trend)	Seasonal variation	Average seasonal variation
2018	1	120					
	2	140					
	3	190			146.25	43.75	51.67
	4	130	580		150.00	−20	−15.4
2019	1	130	590	1 170	156.25	−26.25	−33.3
	2	160	610	1 200	163.75	−3.75	−4.6
	3	220	640	1 250	167.5	52.5	51.67
	4	160	670	1 310	168.75	−8.75	−15.4
2020	1	130	670	1 340	172.5	−42.5	−33.3
	2	170	680	1 350	176.25	−6.25	−4.6
	3	240	700	1 380	181.25	58.75	51.67
	4	170	710	1 410	187.5	−17.5	−15.4
2021	1	160	740	1 450	191.25	−31.25	−33.3
	2	190	760	1 500	193.75	−3.75	−4.6
	3	250	770	1 530			
	4	180	780	1 550			

Forecasting using the moving average method

- **The results from Table 2 enable short-term forecasting:**
  - Plot the moving average trend on a time series graph (Table below).
  - Extrapolate the trend for future periods, as short-term projections are more accurate.
  - Read the forecast trend from the graph for the target period (e.g., quarter 2, 2022).
  - Adjust the forecast using the average seasonal variation for that period.
- For quarter 2 in 2022, the forecast equals the extrapolated trend forecast plus the average seasonal variation.
  - **208 000 + (– $4 600) = 203 400**



An evaluation of the moving average method

Benefits	Limitations
Identifies seasonal variations for predictions.	Long-term forecasts are less accurate as they rely solely on past data, which may not reflect future changes.
Accurate for short-term forecasts in stable conditions.	The moving average method ignores qualitative factors, requiring alternative methods for longer-term forecasts.
Highlights average seasonal variations to aid future quarter planning.	-

Qualitative sales forecasting methods

Sales force composite

- Sales representatives maintain contact with retail customers, gaining insight into market trends and demand. Sales forecasts are created by compiling individual estimates from the sales team.
- The **sales force composite method** is quick and inexpensive but may overlook macroeconomic factors and competitor actions. Customers might overestimate future sales to secure better deals.

Delphi method

- The **Delphi method** gathers expert opinions through anonymous questionnaires on future events like demand or technology changes.
- After each round, responses are summarised and shared with the panel, prompting further input.
- This process continues until a consensus is reached, refining extreme views.
- The method, named after **the Oracle of Delphi**, is more accurate than unstructured expert discussions.

Jury of experts

- The jury of experts uses senior managers to develop forecasts based on their expertise.
- It is quicker and cheaper than the Delphi method but lacks external insights into market conditions and consumer trends.

The impact of sales forecasting on business decisions

- Sales forecasting helps businesses make informed decisions and growth plans.
- Accurate forecasts improve decisions across all functions.
- **For example, if forecasts show declining sales, business can make the following decisions:**
  - **Marketing** – Lower prices, boost promotions, expand distribution, develop new products, or enter stronger markets.
  - **Operations** – Avoid excess capacity, minimize inventory, or ensure flexibility to shift products.
  - **Finance** – Secure short-term funds, cut costs, or seek long-term finance for new products.
  - **HR** – Plan flexible contracts, reduce hiring, or prepare for redundancies.
- Sales forecasts should not be the sole basis for decisions **due to their limitations**.
- However, sharp sales shifts **highlight the value of a flexible, adaptable organization capable of quick decisions.**

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